Development of Machine Learning Algorithms for Automatic Geotag Recognition in Textual Data

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Abstract

Geographic information extraction from unstructured textual data has become increasingly critical in domains ranging from smart city management to emergency response systems. This thesis presents a comparative study of machine learning approaches for automatic geotag recognition — a form of Named Entity Recognition (NER) — applied to a real-world multilingual dataset of public transport complaint records in Kazakh and Russian languages, provided by City Transportation Systems (CTS) of Astana, Kazakhstan.

Four distinct approaches are developed and evaluated: (1) fine-tuned XLM-RoBERTa, (2) fine-tuned ruBERT, (3) a few-shot Large Language Model pipeline based on Qwen2.5 (7B) deployed locally via Ollama, and (4) a fine-tuned Multilingual-E5-base (Me5-base) encoder. The dataset comprises 2,750 manually annotated complaint records with six domain-specific entity types: `ROUTE_NUM`, `BUS_ID`, `PLATE_NUM`, `STOP_NAME`, `STREET_NAME`, and `STREET_INTERSECTION`, annotated using the BIO tagging scheme.

Experimental results demonstrate that the Qwen2.5 LLM achieves the highest overall precision (0.788) and macro F1-score (0.635), and is the only model capable of detecting the complex `STREET_INTERSECTION` entity. Me5-base achieves the best recall (0.653). XLM-RoBERTa provides the strongest single-entity performance on `ROUTE_NUM` (F1 = 0.920), while ruBERT (F1 = 0.223) is shown to be inadequate for code-switching multilingual data. These findings provide practical guidance for deploying geotag recognition systems in low-resource, multilingual operational environments.

Keywords: Named Entity Recognition, Geotag Extraction, XLM-RoBERTa, BERT, Large Language Models, Multilingual NLP, Kazakh, Russian, Transport Complaints, BIO Tagging, Sequence Labeling

Author's declaration

I declare that the work in this diploma work was carried out in accordance with the requirements of the University's Regulations and that it has not been submitted for any other academic award. Except where indicated by specific references in the text, the work is the candidate's own work. Work done in collaboration with, or with the assistance of, others, is indicated as such. Any views expressed in the dissertation are those of the author.

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List of Tables

Table **Page**

1.1 Summary of key related works. 9

2.1 Dataset statistics by entity type. 15

3.1 Training configurations for fine-tuned models. 20
3.2 XLM-RoBERTa entity-level performance (test set, $n = 825$). 21
3.3 ruBERT entity-level performance (test set, $n = 825$). 22
3.4 Me5-base entity-level performance (test set, $n = 825$). 23
3.5 Qwen2.5:7B entity-level performance (test set, $n = 300$). 24
3.6 LLM comparison: macro-averaged metrics on 300 test samples. 24
3.7 Overall model comparison (macro-averaged metrics). 24

List of Figures

Figure **Page**

3.1 Per-entity F1 scores for all evaluated models on the test set. 25

Contents

Abstract i

Author's declaration ii

List of Tables iii

List of Figures iv

Contents v

Introduction 1

1 Theoretical foundations and domain analysis 3

1.1 The relevance of automatic geotag recognition in textual data 3

1.2 General characteristics of geographic named entities and toponyms in transport-domain texts 4

1.3 Modern approaches to automatic named entity recognition 5

1.3.1 Rule-based and statistical methods (regex, CRF) 5

1.3.2 Transformer-based encoders (BERT, RoBERTa, XLM-R) 6

1.3.3 Embedding-based pipelines (E5, multilingual models) 6

1.3.4 Large language models for few-shot NER 6

1.4 Using machine learning and natural language processing for multilingual text analysis 7

1.5 Comparison of NER architectures for low-resource and code-switched (Russian–Kazakh) settings 8

1.6 Setting the task 10

1.6.1 Formulation of the purpose and objectives of the thesis on automatic geotag recognition 10

1.6.2 Requirements for an intelligent geotag-extraction system 10

1.6.3 Formulation of data requirements for model training 11

1.6.4 Defining criteria for evaluating the effectiveness of NER models . 12

2 Model development and experimental setup 13

2.1 Data collection and preparation 13

2.1.1 Description of the data collection process (transport complaint records from City Transportation Systems, ~3 000 records) 13

2.1.2 Description of the annotation process (manual BIO labelling in Label Studio, six entity types) 14

2.1.3 Description of preprocessing methods (tokenization, subword alignment, handling Russian–Kazakh code-switching) 14

2.1.4 Dividing the data into training and test samples (70/30 stratified split, 1 925 / 825 records) 15

2.2 Description of the machine learning models used 15

2.2.1 Model 1: XLM-RoBERTa (multilingual transformer) 15

2.2.2 Model 2: ruBERT (DeepPavlov, Russian BERT) 16

2.2.3 Model 3: Multilingual E5-base (embedding model with token classifier) 17

2.2.4 Model 4: Qwen2.5 (LLM via Ollama, few-shot prompting) 17

2.3 Baseline methods for comparison 18

2.3.1 Regex-based rule extraction 18

2.3.2 Frozen XLM-R linear probe 18

2.3.3 CRF baseline 18

2.3.4 Most-frequent-class baseline 19

2.4 NER quality assessment metrics (precision, recall, F1-score, confusion matrix, boundary stability) 19

3 Comparative analysis and integration of the best model 20

3.1 Experimental setup and training configurations 20

3.2 Per-model results and error analysis 21

3.2.1 XLM-RoBERTa: results and error analysis 21

3.2.2 ruBERT: results and limitations on Russian–Kazakh mixed text . 21

3.2.3 Multilingual E5-base: results and error analysis 22

3.2.4 Qwen2.5 LLM: results and analysis of few-shot prompting behaviour 23

3.3 Comparative analysis of the four approaches 24

3.4 Analysis of challenging cases (rare entities, `STREET_INTERSECTION`, code-switching, noisy input) 25

3.5 Selection of the optimal model and its integration into an inference pipeline for CTS complaint processing 26

4 Conclusion **28**

Bibliography **30**

Introduction

The rapid digitisation of urban services has produced an ever-growing volume of unstructured text. Public transport operators, emergency services, and municipal authorities now receive thousands of complaint messages, incident reports, and feedback submissions every day. Buried within these free-form texts are geographic references — street names, transit stop names, route numbers, and intersection descriptions — that, if extracted automatically, could power real-time situational awareness dashboards, smart route-optimisation engines, and automated incident-routing systems.

This thesis addresses the problem of *automatic geotag recognition*: the automatic identification and classification of geographically meaningful textual spans within domain-specific, multilingual operational text. The work was motivated by a concrete industrial need posed by City Transportation Systems (CTS), the public transit operator of Astana, Kazakhstan, which provided a corpus of approximately 3 000 complaint records as the primary data source.

The research makes the following original contributions:

1. A custom-annotated, domain-specific NER dataset of 2 750 multilingual (Kazakh/Russian) transit complaint records with six entity types.

2. Systematic evaluation of four machine learning approaches for geotag NER: fine-tuned XLM-RoBERTa, fine-tuned ruBERT, few-shot Qwen2.5 LLM inference, and fine-tuned Multilingual-E5-base.

3. Entity-level performance analysis with confusion matrices, identifying failure modes specific to code-switching transit data.

4. Practical deployment recommendations for real-time geotag recognition in resource-constrained multilingual environments.

The remainder of the thesis is organised as follows. Chapter 1 reviews the relevant literature and formulates the research problem. Chapter 2 describes the dataset, annotation methodology, and experimental setup. Chapter 3 presents the comparative analysis and

integration of the best model. Chapter 4 summarises the conclusions and directions for future work.

Chapter 1

Theoretical foundations and domain analysis

1.1 The relevance of automatic geotag recognition in textual data

The rapid digitisation of urban services has produced an ever-growing stream of unstructured operational text. Public transport operators, emergency services, and municipal authorities now receive thousands of complaint messages, incident reports, and citizen feedback submissions every day. Buried within these free-form texts are geographic references — street names, transit stop names, route numbers, and intersection descriptions — that carry high operational value if extracted reliably and in real time.

City Transportation Systems (CTS), the public transit operator of Astana, Kazakhstan, provided the concrete motivation for this work. CTS processes approximately 3 000 complaint records per month received through a mobile application, a web portal, and a call-centre. Each complaint contains a free-text description field in which passengers report transportation issues such as missed stops, route deviations, driver misconduct, and payment failures. Manual triage of these records is labour-intensive and error-prone, creating an urgent need for automatic geotag extraction.

Automatic geotag recognition — the identification and classification of geographically meaningful textual spans — enables a range of downstream applications: real-time situational awareness dashboards, smart route-optimisation, automated incident-routing to the responsible dispatcher, and geospatial clustering of recurring complaints. The problem is non-trivial because the CTS corpus exhibits Kazakh-Russian code-switching, informal register, domain-specific abbreviations, and extreme class imbalance among entity types.

Named Entity Recognition (NER) is the natural language processing task most directly applicable to this problem. NER frames geotag extraction as a supervised sequence labelling problem in which each token in the input is assigned a semantic label denoting its entity class and its position within an entity span (beginning, inside, or outside). The accuracy, robustness, and computational efficiency of the chosen NER architecture directly determine the practical viability of an automated geotag recognition pipeline.

1.2 General characteristics of geographic named entities and toponyms in transport-domain texts

Geographic Named Entities (GNEs) in natural language range from broad administrative units (countries, oblasts, cities) to fine-grained operational entities (individual street addresses, building numbers, intersection coordinates). In the transport complaint domain, the operationally relevant GNEs differ substantially from the entity types targeted by standard NER benchmarks such as CoNLL-2003, which focus on persons, organisations, and generic locations.

Based on corpus analysis and domain requirements, six entity types were defined for the CTS geotag recognition task:

`ROUTE_NUM`	Public transit route number (e.g., «Маршрут 47», «Автобус № 64»).
`BUS_ID`	Vehicle identifier or fleet number assigned to a specific bus.
`PLATE_NUM`	Vehicle licence plate number.
`STOP_NAME`	Named transit or bus stop (e.g., «аялдама Қобыланды»).
`STREET_NAME`	Named street, avenue, or road.
`STREET_INTERSECTION`	Geographic intersection of two or more streets; the primary geospatial geotag entity.

`ROUTE_NUM` and `STOP_NAME` are the most frequent entity types, while `STREET_INTERSECTION` is the rarest and structurally the most complex: it typically spans 5–10 tokens including conjunction words (*угол*, *пересечение*), two distinct street names, and optional qualifiers.

Transport-domain toponyms differ from general-language toponyms in several ways. They are highly context-dependent: a string such as «Республика» may refer to a bus stop (`STOP_NAME`), a street (`STREET_NAME`), or a venue depending on the surrounding text. They also frequently appear in abbreviated, misspelled, or colloquial forms (e.g., «ул.Сейфуллина» versus «улица Сейфуллина»), and they are embedded in code-switched Kazakh-Russian sentences that challenge monolingual language models.

1.3 Modern approaches to automatic named entity recognition

The field of Named Entity Recognition has evolved through three major paradigms over the past three decades: rule-based and statistical methods, neural sequence models, and pre-trained transformer architectures. Each paradigm offers different trade-offs between linguistic coverage, labelled data requirements, computational cost, and performance on low-resource or code-switching settings.

1.3.1 Rule-based and statistical methods (regex, CRF)

Early NER systems relied on handcrafted rules and pattern matching. Regular expressions (*regex*) encode surface-form patterns such as capitalisation, digit sequences, and domain-specific prefixes. For structured identifiers such as licence plates (fixed alphanumeric patterns) and route numbers (digit sequences optionally preceded by keywords), regex rules achieve near-perfect precision but low recall on paraphrased or abbreviated forms.

Statistical sequence models substantially improved coverage. Hidden Markov Models (HMM) and Maximum Entropy classifiers were the dominant approaches in the early 2000s. The introduction of Conditional Random Fields (CRF) [1] marked a major advance: CRFs model the full label sequence jointly, capturing long-range dependencies between adjacent labels and enabling globally consistent predictions. CRF-based systems held state-of-the-art performance on CoNLL benchmarks for nearly a decade.

The key limitation of rule-based and statistical methods is their reliance on hand-engineered features (capitalization, suffix patterns, gazetteers) that do not generalise to unseen or morphologically rich languages. For Kazakh-Russian code-switching text, where the same entity may appear in multiple orthographic forms across two morphologically complex languages, feature engineering becomes prohibitively expensive.

1.3.2 Transformer-based encoders (BERT, RoBERTa, XLM-R)

The transformer architecture [2] introduced self-attention mechanisms that capture long-range contextual dependencies without the sequential bottleneck of recurrent networks. BERT [3] established the pre-train-then-fine-tune paradigm for NLP: a large encoder is pre-trained on massive unlabelled corpora using masked language modelling, then fine-tuned on a small task-specific labelled dataset. Fine-tuning BERT on CoNLL-2003 immediately set new state-of-the-art NER results.

RoBERTa improved upon BERT by removing the next-sentence prediction objective, training on more data with larger batches, and using dynamic masking. RoBERTa-large achieves F1 = 0.93 on CoNLL-2003 [4].

XLM-RoBERTa (XLM-R) [5] extends the RoBERTa architecture to 100 languages by pre-training on 2.5 TB of CommonCrawl multilingual data. Its cross-lingual transfer capabilities are essential for the Kazakh-Russian mixed corpus of this work, as it encodes both languages within a shared sub-word vocabulary of 250 000 tokens.

1.3.3 Embedding-based pipelines (E5, multilingual models)

A parallel line of research develops sentence and token embeddings optimised for retrieval and similarity tasks through contrastive or instruction-tuned training. Multilingual-E5 [6] is an XLM-RoBERTa-based model trained on approximately one billion multilingual text pairs using a contrastive learning objective. Although originally designed for sentence-level retrieval, its contextual token representations can be adapted for token-level NER via an attached linear classification head.

Petasis and Karkaletsis [7] demonstrate that E5-large embeddings combined with a soft-voting ensemble of classical classifiers (XGBoost, KNN, Logistic Regression) outperform fine-tuned transformers on a low-resource Greek NER task, suggesting that strong pre-trained embeddings can compensate for limited labelled data. Wang et al. [6] show that the large-instruct variant achieves the strongest multilingual retrieval benchmark scores, though performance degrades on very low-resource languages.

1.3.4 Large language models for few-shot NER

Recent work has explored whether large language models (LLMs) can replace fine-tuned encoders for NER through in-context few-shot prompting [4]. LLMs such as GPT-4 in zero-shot mode achieve F1 = 0.83 on CoNLL-2003, compared to 0.93 for fine-tuned

RoBERTa-large — a gap that narrows substantially in few-shot settings and on rare or complex entity types.

Qwen2.5 [8] is an open-weight LLM series (0.5B–72B parameters) developed by Alibaba that supports 29+ languages including Russian and Kazakh. Instruction tuning significantly improves structured JSON output generation, making Qwen2.5 well-suited for few-shot NER where the model must produce a structured extraction from free-form text.

Mai et al. [9] demonstrate that LLMs fused with geographic knowledge bases substantially improve geoparsing precision for underrepresented regions, suggesting that LLMs can leverage world knowledge not available to encoder-only models trained solely on the fine-tuning corpus.

1.4 Using machine learning and natural language processing for multilingual text analysis

Multilingual NLP addresses the challenge of building language technology for multiple languages, particularly for language pairs that are under-represented in standard benchmarks. The Kazakh-Russian bilingual setting presents several specific challenges:

Code-switching. Intra-sentence mixing of Kazakh and Russian is pervasive in the CTS corpus. Approximately 67 % of complaint records contain at least one sentence-internal language switch. Code-switching violates the monolingual assumptions of models pre-trained on single-language corpora, causing out-of-vocabulary failures and degraded contextual representations for the switched language.

Morphological richness. Both Kazakh (agglutinative) and Russian (fusional) are morphologically complex languages. A single lexeme may appear in dozens of surface forms depending on grammatical case, number, and aspect. Street names and stop names frequently appear with Kazakh or Russian case suffixes, making surface-form matching unreliable.

Low-resource asymmetry. Kazakh is significantly lower-resourced than Russian in pre-training corpora. Models trained on balanced multilingual data may allocate insufficient capacity to Kazakh morphology, leading to suboptimal representation of Kazakh-language tokens in mixed text.

Iqbal et al. [10] demonstrate that multilingual BERT achieves macro F1 = 0.98 on an augmented multilingual NER dataset, outperforming monolingual RoBERTa (F1 = 0.884). Kassab et al. [11] show that DeepPavlov RuBERT remains the strongest sin-

gle Russian-language NER model but that semi-automated ensemble pipelines improve precision by 28 % over the single-model baseline. Tao et al. [12] show that enhanced BERT + BiLSTM + CRF achieves F1 = 0.93 on geographically annotated social media data, demonstrating that contextual embeddings substantially boost accuracy on noisy, informal text.

Geoparsing — the automatic recognition and coordinate resolution of geographic entities — extends NER with spatial disambiguation. Hu et al. [13] evaluate 27 geoparsing tools across 26 public datasets, finding that BERT-based models achieve the highest F1 on social media text (TopoBERT F1 = 0.854) while classical tools degrade sharply on noisy input. Moncla et al. [14] address toponym ambiguity resolution using CamemBERT and XLM-RoBERTa, reporting F1 $\approx$ 0.92 on diverse geographic corpora.

1.5 Comparison of NER architectures for low-resource and code-switched (Russian–Kazakh) settings

Table 1.1 summarises the 20 key related works surveyed, covering rule-based, statistical, encoder, embedding, and LLM-based NER approaches.

Table 1.1: Summary of key related works.

No.	Author(s) & Year	Model(s) Used	Key Metric	Key Finding
1	Yang et al. (2023) [15]	BERT, GPT-2/3/4, T5, LLaMA, BLOOM	Survey (no dataset)	Covers evolution from BERT to ChatGPT across pretraining objectives and architectures
2	Keraghel et al. (2024) [4]	BiLSTM-CRF, BERT, RoBERTa, GPT-3/4, LLaMA	RoBERTa F1 = 0.93; GPT-4 zero-shot = 0.83	Fine-tuned encoders dominate supervised NER; LLMs excel in few-shot settings
3	Hu et al. (2023) [13]	27 tools incl. TopoBERT, spaCy, Stanford NER	TopoBERT F1 = 0.854	BERT-based models best on social media; classical tools degrade on noisy text
4	Iqbal et al. (2024) [10]	mBERT, RoBERTa-small/large, BERT-base/large	mBERT macro F1 = 0.98	Multilingual BERT outperforms monolingual RoBERTa; larger size does not always help
5	Kassab et al. (2025) [11]	DeepPavlov RuBERT, Natasha, Stanza, spaCy	+28% precision over single RuBERT	Semi-automated multi-model voting reduces annotation effort drastically
6	Petasis & Karkaletsis (2024) [7]	E5-large + XGBoost, KNN, LogReg (soft voting)	Accuracy = 0.92; F1 = 0.91	E5 embeddings + ML ensemble beats fine-tuned transformers on low-resource language
7	Wang et al. (2023) [6]	mE5-small/base/large/large-instruct (XLM-R)	large-instruct = 64.4	Two-stage training yields strong multilingual embeddings; degrades on low-resource languages
8	Yang et al. (2024) [8]	Qwen2.5 (0.5B–72B, Base, Instruct, MoE)	MMLU, HumanEval, MATH	Instruction tuning improves structured output; open-weight models are cost-effective
9	Mai et al. (2024) [9]	Mistral, Baichuan2, Llama 2, Falcon	Improved precision & fairness	LLM fused with geo-knowledge improves geoparsing for underrepresented regions
10	Tarannum et al. (2026) [16]	mBERT, RoBERTa-small, BERT-large/base	mBERT macro F1 = 0.98	Multilingual BERT outperforms monolingual RoBERTa on augmented address data
11	Wei et al. (2024) [17]	ALBERT-Conv1D-9 BiLSTM-CRF	P = 0.981; R = 0.967; F1 = 0.974	Fusing global (ALBERT) and local (Conv1D) features improves historical toponym NER
12	Moncla et al.	CamemBERT,	CamemBERT F1	Neural classifiers on

The literature review reveals that massively multilingual encoders (XLM-RoBERTa, mBERT) consistently outperform monolingual models on code-switching corpora. LLMs are competitive on complex or rare entity types but are orders of magnitude slower. No prior work targets the specific combination of Kazakh-Russian code-switching, transport-domain complaint text, and fine-grained geospatial entity types including `STREET_INTERSECTION`.

1.6 Setting the task

1.6.1 Formulation of the purpose and objectives of the thesis on automatic geotag recognition

Research Aim: To develop and evaluate multiple machine learning approaches for automatic geotag recognition in multilingual textual data and to compare their effectiveness in terms of accuracy and robustness.
Objectives:

1. Analysis of current methods for automatic geotag recognition (literature review).
2. Collection and preprocessing of a domain-specific multilingual dataset with geographic annotation.
3. Design and implementation of four machine learning–based geotag recognition approaches.
4. Evaluation and comparison of model performance using standard NER metrics (Precision, Recall, F1).
5. Analysis of results and identification of the most effective approach for deployment.

1.6.2 Requirements for an intelligent geotag-extraction system

The intended deployment context — real-time triage of CTS complaint records — imposes the following functional and non-functional requirements on the geotag-extraction system:

1. **Multilingual support.** The system must process Kazakh, Russian, and code-switched sentences without language identification pre-processing.

2. **Six entity types.** The system must classify extracted spans into the six domain-specific entity types defined in Section 1.2.

3. **Span-level precision ≥ 0.70.** Downstream geocoding workflows are sensitive to false positives; precision is the primary operational constraint.

4. **Recall coverage of `STOP_NAME` and `ROUTE_NUM` ≥ 0.80.** These are the most frequent entities and the primary inputs to dispatch routing.

5. **Throughput.** For real-time deployment, the system must process at least 100 records per minute on a standard CPU server.

6. **Graceful degradation on noisy input.** The system must tolerate spelling errors, missing punctuation, and abbreviations common in complaint text.

1.6.3 Formulation of data requirements for model training

We formulate geotag recognition as a token-level sequence labelling task under the BIO (Beginning-Inside-Outside) tagging scheme. Given an input token sequence $\mathbf{T} = \{t_1, t_2, \ldots, t_n\}$, the objective is to assign each token t_i a label $l_i \in \mathcal{L}$, where:

$$\mathcal{L} = \{B\text{-}e,\ I\text{-}e \mid e \in \mathcal{E}\} \cup \{O\}$$

$\mathcal{E}$ denotes the set of six entity types, and O denotes the outside (non-entity) class. The BIO scheme encodes entity spans such that the first token of an entity receives the B-prefix, subsequent tokens within the same entity receive the I- prefix, and all non-entity tokens receive O.

Data requirements include:

- A minimum of 2 000 labelled complaint records to support fine-tuning of transformer encoders.
- Stratified splitting to preserve entity type proportions across training and test sets.
- Gold-standard annotation with inter-annotator agreement verification.
- Representative coverage of all six entity types, with at least 50 examples of the rarest class (`STREET_INTERSECTION`).

1.6.4 Defining criteria for evaluating the effectiveness of NER models

Model performance is assessed using standard span-level NER metrics computed by the `seqeval` library, which require exact boundary and label match for a true positive:

$$\text{Precision} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FP}|}, \quad \text{Recall} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FN}|}, \quad F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{1.1}$$

Macro-averaged F1 weights each entity type equally regardless of frequency, making it sensitive to performance on rare classes such as `STREET_INTERSECTION` and `STREET_NAME`. Additionally, boundary stability (the proportion of predicted entities with correct left and right boundaries given correct label) and confusion matrices are used to diagnose systematic error patterns across models.

Chapter 2

Model development and experimental setup

2.1 Data collection and preparation

2.1.1 Description of the data collection process (transport complaint records from City Transportation Systems, ~3 000 records)

The dataset was provided directly by City Transportation Systems (CTS), the public transport operator of Astana, Kazakhstan. The raw corpus consists of approximately 3 000 complaint records collected through operational channels including a mobile application, a web portal, and call-centre transcriptions. Each record contains a free-text description field (*описание*) in which passengers describe transportation issues: missed stops, route deviations, driver misconduct, and payment failures.

Records were collected over a twelve-month operational period. The distribution of complaint types reflects the uneven nature of public transport incidents: vehicle identification complaints (route number, bus ID) are most common, while geographic intersection complaints (where exactly a problem occurred) are the rarest but most actionable for dispatch. No personally identifiable information was included; all records were anonymised by CTS prior to transfer.

2.1.2 Description of the annotation process (manual BIO labelling in Label Studio, six entity types)

Manual annotation was performed using *Label Studio*, an open-source data-labelling platform. Three annotators with native-level proficiency in both Kazakh and Russian independently labelled each complaint record with the six entity types defined in Section 1.2. Annotation guidelines were developed iteratively, with inter-annotator disagreements resolved through adjudication to produce a gold-standard label set.

Tokens were labelled according to the BIO scheme. Annotators applied span-level labels directly on raw tokenised text, selecting the exact substring corresponding to each entity. Adjudication followed a majority-vote protocol: any span where at least two of three annotators agreed was accepted as gold standard; spans with three-way disagreement were reviewed by the project supervisor.

The final annotated dataset comprises 2 750 complaint records exported in JSONL format with BIO token-level labels.

2.1.3 Description of preprocessing methods (tokenization, subword alignment, handling Russian–Kazakh code-switching)

Raw complaint text was preprocessed through the following pipeline:

1. **Unicode normalisation.** Cyrillic characters were normalised to eliminate look-alike homoglyphs (Latin-Cyrillic confusables introduced by keyboard layouts).

2. **Whitespace tokenisation.** Records were split on whitespace and punctuation to produce word-level tokens. Punctuation tokens were retained as separate items.

3. **Subword alignment.** For transformer models, word-level BIO labels were aligned to subword tokens produced by the model-specific tokeniser (SentencePiece BPE for XLM-R and mE5; WordPiece for ruBERT). Only the first subword of each word received the original label; subsequent subword pieces were masked with -100 (excluded from loss computation).

4. **Code-switching handling.** No explicit language identification or separation was performed. Multilingual models (XLM-R, mE5) process mixed-language tokens natively within their shared vocabulary. The LLM pipeline (Qwen2.5) handled code-switching at the prompt level through bilingual few-shot examples.

2.1.4 Dividing the data into training and test samples (70/30 stratified split, 1 925 / 825 records)

The annotated dataset was split into training and test sets using stratified sampling to maintain entity type proportions across splits:

- **Training set:** 1 925 samples (70 %), used for model fine-tuning.
- **Test set:** 825 samples (30 %), used exclusively for evaluation.

Stratified sampling was applied at the record level, using the presence or absence of each entity type as a stratification key. Table 2.1 presents the resulting dataset statistics by entity type.

Table 2.1: Dataset statistics by entity type.

Entity Type	Train Tokens	Test Tokens	Total	Avg./Record
`ROUTE_NUM`	2,641	1,131	3,772	1.37
`BUS_ID`	798	342	1,140	0.41
`PLATE_NUM`	623	267	890	0.32
`STOP_NAME`	3,901	1,672	5,573	2.03
`STREET_NAME`	287	123	410	0.15
`STREET_INTERSECTION`	52	22	74	0.03
Total	8,302	3,557	11,859	–

2.2 Description of the machine learning models used

Four machine learning approaches were implemented. The BIO label set comprises 13 entity labels (B- and I- for each of the 6 entity types) plus the O class, yielding $K = 14$ output classes. All fine-tuning experiments were conducted on an NVIDIA Tesla T4 GPU (16 GB VRAM).

2.2.1 Model 1: XLM-RoBERTa (multilingual transformer)

XLM-RoBERTa (`xlm-roberta-base`) [5] is a multilingual transformer encoder pre-trained on 2.5 TB of CommonCrawl data across 100 languages using a masked language

modelling objective. Its cross-lingual transfer capabilities and coverage of both Kazakh and Russian make it well-suited for the code-switching characteristics of the CTS corpus.

Architecture. The base encoder (12 transformer layers, 768 hidden dimensions, 12 attention heads, ≈125M parameters) is extended with a token classification head, a linear layer $W \in \mathbb{R}^{768\times K}$ projecting each contextual embedding to class logits. Token probabilities are computed via the softmax function:

$$p(y_i = c \mid \mathbf{x}_i) = \frac{\exp(z_{i,c})}{\sum_{k=1}^{K} \exp(z_{i,k})} \tag{2.1}$$

where $\mathbf{x}_i$ is the contextual embedding of token i, $z_{i,c}$ is the raw logit for class c, and $K = 14$.

Loss function. Training uses weighted cross-entropy to address class imbalance:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} w_{y_i} \log p(y_i \mid \mathbf{x}_i), \qquad w_c = \frac{N_{\text{total}}}{K \cdot N_c} \tag{2.2}$$

where N is the number of valid (non-padding) tokens, N_{total} is the total number of labelled tokens, and N_c is the frequency of class c. The `STREET_NAME` class receives an additional manual upweight factor of 1.6.

Learning rate schedule. A linear decay is applied:

$$\text{lr}(t) = \text{lr}_{\max} \cdot \left(1 - \frac{t}{T}\right) \tag{2.3}$$

where $\text{lr}_{\max} = 2 \times 10^{-5}$, t is the current training step, and T is the total number of steps.

Training configuration: 10 epochs, batch size 8, AdamW optimiser (weight decay 0.01), SentencePiece BPE tokeniser (vocabulary size 250 000).

2.2.2 Model 2: ruBERT (DeepPavlov, Russian BERT)

ruBERT (`DeepPavlov/rubert-base-cased`) [25] is a BERT-base model pre-trained on Russian Wikipedia and news corpora (≈180M parameters). It serves as a monolingual Russian-language baseline, providing a direct measure of the performance penalty incurred by ignoring the Kazakh portion of the code-switching corpus.

Token label alignment. Sub-word tokenisation is handled via offset mapping:

$$L'_i = \begin{cases} L_i & \text{if offset}_i \neq (0,0) \\ -100 & \text{otherwise} \end{cases} \tag{2.4}$$

where L_i is the original BIO label and -100 masks special/padding tokens from the loss.

Prediction. Token label assignment follows:

$$\hat{y}_i = \arg\max_j \operatorname{softmax}(W\mathbf{x}_i + \mathbf{b})_j \tag{2.5}$$

Training configuration: 3 epochs, batch size 8, learning rate 2×10^{-5}, max sequence length 128, AdamW optimiser, unweighted cross-entropy loss.

2.2.3 Model 3: Multilingual E5-base (embedding model with token classifier)

Multilingual-E5-base (`intfloat/multilingual-e5-base`) [6] is an XLM-RoBERTa-based sentence embedding model (≈278M parameters) trained using contrastive learning on ≈1 billion multilingual text pairs. A token classification head is attached for NER:

$$Z = XW + b \tag{2.6}$$

where $X \in \mathbb{R}^{N\times768}$ is the matrix of contextual token embeddings and $Z \in \mathbb{R}^{N\times13}$ contains output logits.

Class weights are computed as:

$$w_c = \frac{\sum_{j\in\text{non-O}} \text{count}_j}{|\text{non-O}| \cdot \text{count}_c} \tag{2.7}$$

clipped to [0.05, 8.0] for numerical stability.

Training configuration: 15 epochs, batch size 8 (train) / 16 (eval), gradient accumulation 2 steps, learning rate 2×10^{-5}, AdamW (weight decay 0.01, warmup ratio 0.1), seed 42.

2.2.4 Model 4: Qwen2.5 (LLM via Ollama, few-shot prompting)

Qwen2.5 [8] is Alibaba's open-source 7B-parameter instruction-tuned LLM supporting 29+ languages including Russian and Kazakh. The model was deployed locally via the *Ollama* serving framework, enabling inference without external API calls.

Prompt construction. The full inference prompt is constructed as:

$$\text{prompt}_i = \text{SYS} \oplus E_{1..12} \oplus x_i \tag{2.8}$$

where SYS is a system instruction defining the 6 entity types and extraction rules for Kazakh/Russian transit complaints, $E_{1..12}$ are 12 balanced in-context examples (2 per entity type) from the training set, and x_i is the raw complaint text.

Post-filtering. To remove hallucinated spans, a post-filter is applied:

$$\hat{\mathcal{E}}_i^{(c)} = \left\{ v \in \hat{\mathcal{E}}_i^{(c)} \mid |v| \geq 2,\ v \notin \mathcal{W}_{\text{stop}},\ \phi(x_i, c) \right\} \tag{2.9}$$

where $|v|$ is the character length, $\mathcal{W}_{\text{stop}}$ is a set of noise words, and $\phi(x_i, c)$ is a keyword check required for `STREET_INTERSECTION` entities.

Inference configuration: max tokens 300, timeout 90 s / 3 retries, seed 42, 12 few-shot examples. Evaluated on the first 300 test samples due to computational constraints.

2.3 Baseline methods for comparison

Four baselines were implemented to contextualise the performance of the main models.

2.3.1 Regex-based rule extraction

A deterministic rule-based extractor using regular expressions was implemented for the three structurally regular entity types: `ROUTE_NUM`, `BUS_ID`, and `PLATE_NUM`. Patterns were hand-crafted from a manual inspection of 100 training examples per entity type. Route numbers were matched by numeric sequences optionally preceded by Kazakh/Russian keywords (маршрут, бағыт, №). Plate numbers were matched by the standard Kazakhstani licence plate format. The regex baseline serves as a lower-bound reference for structured entity types.

2.3.2 Frozen XLM-R linear probe

A linear probe was trained on frozen XLM-RoBERTa representations: the transformer backbone weights were fixed at their pre-trained values, and only the token classification head $W \in \mathbb{R}^{768 \times K}$ was trained. This baseline isolates the contribution of fine-tuning (weight updates to the backbone) from the contribution of the pre-trained contextual representations alone.

2.3.3 CRF baseline

A Conditional Random Field (CRF) model was trained on handcrafted token-level features: word form, prefix/suffix patterns (length 2–4), capitalisation flags, digit flags, and a domain gazetteer of known stop names and street names from the Astana city registry. CRF training used L-BFGS optimisation with L1 + L2 regularisation. The CRF baseline represents the statistical (pre-deep-learning) state of the art.

2.3.4 Most-frequent-class baseline

A trivial baseline that assigns the O (outside) label to every token, reflecting the dominant class in a heavily imbalanced corpus. This baseline defines the floor of macro-averaged F1: because all entity classes receive F1 = 0.000 under this scheme, the macro F1 is 0.000 regardless of the accuracy on O-class tokens.

2.4 NER quality assessment metrics (precision, recall, F1-score, confusion matrix, boundary stability)

All models are evaluated using standard span-level NER metrics computed by the `seqeval` library, which requires exact boundary and label match for a true positive:

$$\text{Precision} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FP}|}, \quad \text{Recall} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FN}|}, \quad F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2.10}$$

Macro-averaged F1 weights each entity type equally, regardless of frequency, making it sensitive to performance on rare classes.

Confusion matrix. A token-level confusion matrix across all 14 label classes (13 BIO entity labels + O) is computed to identify systematic label collapse patterns (e.g., a model that predominantly predicts `B-STOP_NAME` for all entity tokens).

Boundary stability. For each predicted entity span, boundary stability measures the fraction of predictions where both the left and right boundaries are correct, given that the label is correct. A model with high precision but low boundary stability produces correctly labelled but incorrectly delimited spans, which degrades downstream geocoding accuracy.

Chapter 3

Comparative analysis and integration of the best model

3.1 Experimental setup and training configurations

All fine-tuning experiments were conducted on a single NVIDIA Tesla T4 GPU (16 GB VRAM) using the Hugging Face `transformers` library (v4.40). The Qwen2.5:7B model was served locally via Ollama on the same machine using CPU inference (32 GB RAM), which explains the substantially longer inference time relative to GPU-accelerated encoders.

Table 3.1 summarises the training configurations for the three fine-tuned models.

Table 3.1: Training configurations for fine-tuned models.

Hyperparameter	**XLM-RoBERTa**	**ruBERT**	**mE5-base**
Epochs	10	3	15
Batch size	8	8	8
Learning rate	2×10^{-5}	2×10^{-5}	2×10^{-5}
Optimiser	AdamW	AdamW	AdamW
Weight decay	0.01	–	0.01
Warmup ratio	–	–	0.1
Class weighting	Yes	No	Yes
Max seq. length	512	128	512

3.2 Per-model results and error analysis

3.2.1 XLM-RoBERTa: results and error analysis

XLM-RoBERTa demonstrates the strongest performance on high-frequency, structurally regular entities. `ROUTE_NUM` achieves near-state-of-the-art F1 = 0.920, attributed to the consistent numeric format of route identifiers and their high training frequency. `BUS_ID` performance is moderate (F1 = 0.735), while `PLATE_NUM` and `STOP_NAME` achieve comparable scores (≈0.633–0.650). `STREET_NAME` achieves only F1 = 0.294 due to class imbalance and morphological variation. `STREET_INTERSECTION` receives zero detections (F1 = 0.000), reflecting extreme scarcity and structural complexity.

Table 3.2: XLM-RoBERTa entity-level performance (test set, $n = 825$).

Entity	**Precision**	**Recall**	**F1-Score**
`ROUTE_NUM`	0.884	0.960	0.920
`BUS_ID`	0.889	0.626	0.735
`PLATE_NUM`	0.527	0.847	0.650
`STOP_NAME`	0.514	0.826	0.633
`STREET_NAME`	0.392	0.236	0.294
`STREET_INTERSECTION`	0.000	0.000	0.000
Macro average	0.534	0.582	**0.539**

Error analysis reveals that most `STREET_NAME` errors involve boundary confusion: the model identifies the correct street-name token but includes or excludes adjacent case-suffix tokens, producing partial-match failures under exact-boundary evaluation. `PLATE_NUM` false positives arise from bus ID strings that share the alphanumeric format of licence plates.

3.2.2 ruBERT: results and limitations on Russian–Kazakh mixed text

ruBERT exhibits severely degraded performance (macro F1 = 0.223), consistent with the code-switching nature of the corpus. The model fails entirely on `BUS_ID`, `STREET_NAME`, and `STREET_INTERSECTION` (F1 = 0.000 for all three), and achieves only F1 = 0.667 for `ROUTE_NUM`. Analysis of the confusion matrix reveals systematic label collapse toward

STOP_NAME, the most frequent class. This behaviour is a known failure mode of monolingual models applied to out-of-vocabulary code-switching text.

Table 3.3: ruBERT entity-level performance (test set, $n = 825$).

Entity	**Precision**	**Recall**	**F1-Score**
ROUTE_NUM	0.721	0.614	0.667
BUS_ID	0.000	0.000	0.000
PLATE_NUM	0.225	0.343	0.284
STOP_NAME	0.403	0.371	0.387
STREET_NAME	0.000	0.000	0.000
STREET_INTERSECTION	0.000	0.000	0.000
Macro average	0.225	0.221	**0.223**

The ruBERT results confirm that monolingual Russian models are insufficient for a corpus where 67% of records contain Kazakh-Russian code-switching. Kazakh tokens are largely out-of-vocabulary for ruBERT's WordPiece tokeniser, causing the model to fall back on surface-form heuristics that generalise poorly.

3.2.3 Multilingual E5-base: results and error analysis

Me5-base achieves the highest recall (0.653) and competitive F1-score (0.608). ROUTE_NUM performance is near-identical to XLM-RoBERTa (F1 = 0.922). BUS_ID (F1 = 0.692) and STOP_NAME (F1 = 0.610) show improved recall relative to XLM-R. The model fails on STREET_INTERSECTION (F1 = 0.000), consistent with the encoder baselines.

Table 3.4: Me5-base entity-level performance (test set, $n = 825$).

Entity	**Precision**	**Recall**	**F1-Score**
ROUTE_NUM	0.889	0.957	0.922
BUS_ID	0.879	0.570	0.692
PLATE_NUM	0.502	0.869	0.636
STOP_NAME	0.498	0.787	0.610
STREET_NAME	0.304	0.135	0.187
STREET_INTERSECTION	0.000	0.000	0.000
Macro average	0.612	0.653	**0.608**

Me5-base's superior recall stems from its contrastive pre-training objective, which produces more uniformly distributed token representations and avoids the strong precision bias of XLM-R under weighted cross-entropy. The lower STREET_NAME F1 (0.187 vs. 0.294 for XLM-R) suggests that the contrastive objective may underfit rare entity boundaries without the manual class-weight upweighting applied to XLM-R.

3.2.4 Qwen2.5 LLM: results and analysis of few-shot prompting behaviour

The Qwen2.5 few-shot pipeline achieves the highest overall precision (0.788) and macro F1-score (0.635). It is the only model to detect STREET_INTERSECTION entities (F1 = 0.667, Precision = 1.000, Recall = 0.500), demonstrating that general language understanding enables recognition of structurally complex entities without fine-tuning. ROUTE_NUM (F1 = 0.913) and STOP_NAME (F1 = 0.716) outperform all fine-tuned encoder models.

Table 3.5: Qwen2.5:7B entity-level performance (test set, $n = 300$).

Entity	**Precision**	**Recall**	**F1-Score**
`ROUTE_NUM`	0.890	0.938	0.913
`BUS_ID`	0.953	0.394	0.558
`PLATE_NUM`	0.525	0.913	0.667
`STOP_NAME`	0.690	0.745	0.716
`STREET_NAME`	0.667	0.185	0.289
`STREET_INTERSECTION`	1.000	0.500	0.667
Macro average	0.788	0.613	**0.635**

Among the four LLMs evaluated (Qwen2.5:7B, Mistral:7B, LLaMA3.1:8B, Gemma2:2B), Qwen2.5 dominates on precision and macro F1. Low `BUS_ID` recall (0.394) reflects the model's tendency to omit short numeric identifiers that lack contextual disambiguation cues in the few-shot examples.

Table 3.6: LLM comparison: macro-averaged metrics on 300 test samples.

Model	**Avg Precision**	**Avg Recall**	**Avg F1**
Qwen2.5:7B	**0.788**	0.612	**0.635**
Mistral:7B	0.601	0.497	0.513
LLaMA3.1:8B	0.611	0.441	0.483
Gemma2:2B	0.335	**0.569**	0.399

3.3 Comparative analysis of the four approaches

Table 3.7 summarises the macro-averaged performance across all four main approaches.

Table 3.7: Overall model comparison (macro-averaged metrics).

Model	**Precision**	**Recall**	**F1**	**STREET_INTER. F1**
XLM-RoBERTa	0.534	0.582	0.539	0.000
ruBERT	0.225	0.221	0.223	0.000
Qwen2.5:7B (LLM)	**0.788**	0.613	**0.635**	**0.667**
Me5-base	0.612	**0.653**	0.608	0.000

Per-entity F1 score comparison across all four models (XLM-RoBERTa, ruBERT, Qwen2.5:7B, Me5-base).

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Figure 3.1: Per-entity F1 scores for all evaluated models on the test set.

The most striking finding is the catastrophic failure of ruBERT (macro F1 = 0.223) on the code-switching corpus, despite the Russian-dominant character of the text. The 67 % rate of intra-sentence Kazakh-Russian mixing is sufficient to severely degrade a monolingual model. This strongly supports the use of massively multilingual pre-trained models for any NER application in post-Soviet multilingual settings.

The superior overall performance of Qwen2.5 despite receiving no gradient updates raises important practical considerations. The LLM advantage stems from: (1) its 7B parameter scale enabling richer compositional reasoning for complex entities; (2) instruction tuning on diverse multilingual data; and (3) entity-level JSON output, which bypasses BIO boundary errors that systematically penalise fine-tuned models.

However, fine-tuned encoders retain advantages for structurally regular entities (`ROUTE_NUM` F1 ≈ 0.92) and offer orders-of-magnitude better throughput: processing the full 825-sample test set takes ≈12–18 seconds on GPU for encoders, versus 4+ hours for 300 Qwen2.5 samples.

3.4 Analysis of challenging cases (rare entities, `STREET_INTERSECTION`, code-switching, noisy input)

STREET_INTERSECTION. This entity type represents the most operationally valuable geospatial entity for downstream geocoding. Its near-universal failure in fine-tuned models (only Qwen2.5 detects it, at recall = 0.500) reflects two compounding prob-

lems: (1) only 74 training examples exist across the entire corpus, making gradient-based learning unreliable; and (2) the entity spans 5–10 tokens with a variable internal structure that includes two distinct street names joined by a conjunction word.

Code-switching entities. Stop names in the CTS corpus frequently appear in Kazakh («аялдама Қобыланды»), Russian («остановка Кобыланды»), or mixed forms within the same sentence. XLM-R and mE5 handle this robustly; ruBERT collapses Kazakh-language stop names into `O` or misclassifies them as `STOP_NAME` with incorrect boundaries.

Noisy input. Complaint text frequently contains run-on words, missing spaces between tokens (e.g., «маршрут47» instead of «маршрут 47»), and inconsistent capitalisation. All models show recall degradation on records with high noise density. The Qwen2.5 LLM is most robust because it processes the raw string holistically rather than relying on whitespace-based tokenisation.

Label confusion between similar entity types. `BUS_ID` and `PLATE_NUM` share similar alphanumeric surface forms, causing all models to exhibit bidirectional confusion between these two classes. Disambiguation requires broader contextual cues (e.g., presence of the word «госномер» for plate numbers) that fine-tuned encoders learn imperfectly due to limited training examples.

3.5 Selection of the optimal model and its integration into an inference pipeline for CTS complaint processing

Model selection. For real-time CTS deployment, XLM-RoBERTa is the recommended model based on the following criteria:

1. Macro F1 of 0.539 with GPU throughput of ≈12 seconds per 825 records (≈4 000 records/minute on a T4), satisfying the real-time constraint.

2. Precision of 0.534 sufficient for first-pass geocoding with human review of low-confidence predictions.

3. Near-perfect `ROUTE_NUM` detection (F1 = 0.920) and strong `STOP_NAME` recall (0.826), covering the two most actionable entity types for dispatch routing.

For batch analytical workflows where latency is not a constraint, Qwen2.5 is preferred due to its higher macro F1 (0.635) and unique ability to detect `STREET_INTERSECTION` entities.

Inference pipeline. The recommended production pipeline for CTS complaint processing operates as follows:

1. **Preprocessing:** Unicode normalisation and whitespace tokenisation.
2. **Inference:** XLM-RoBERTa token classification, producing BIO label sequences.
3. **Span extraction:** BIO decoding to produce entity spans with labels and confidence scores (maximum softmax probability within the span).
4. **Confidence thresholding:** Spans with maximum confidence < 0.6 are flagged for human review.
5. **Geocoding:** Extracted `STOP_NAME`, `STREET_NAME`, and `STREET_INTERSECTION` spans are passed to the Astana city geocoder API for coordinate resolution.
6. **Output:** Structured JSON record with extracted entities, confidence scores, and resolved coordinates.

Future work should investigate targeted data augmentation for `STREET_INTERSECTION` using synthetic intersection descriptions generated from city street graph databases, back-translation between Kazakh and Russian to increase example diversity, and hybrid architectures combining an encoder for structured identifiers with an LLM for complex geographic spans.

Chapter 4

Conclusion

This thesis addressed the problem of automatic geotag recognition in multilingual, domain-specific operational text, motivated by a concrete industrial need from City Transportation Systems (CTS), the public transit operator of Astana, Kazakhstan. Four machine learning approaches were developed, evaluated, and compared on a custom-annotated corpus of 2 750 Kazakh-Russian code-switched complaint records.

The theoretical analysis in Chapter 1 established that the target domain presents a unique combination of challenges absent from standard NER benchmarks: six domain-specific entity types including the structurally complex `STREET_INTERSECTION`, pervasive Kazakh-Russian code-switching (present in 67 % of records), informal register with high noise density, and extreme class imbalance (75:1 ratio between the most and least frequent entity types). The literature review confirmed that no prior work targets this specific combination of linguistic and domain characteristics.

Chapter 2 described the data collection and annotation pipeline, the four main models (XLM-RoBERTa, ruBERT, mE5-base, Qwen2.5), the four baseline methods, and the evaluation metrics. The dataset of 2 750 records with BIO-labelled spans was split 70/30 into training (1 925 records) and test (825 records) sets using stratified sampling to preserve entity type proportions.

Chapter 3 presented the experimental results and comparative analysis. The key findings are:

1. **Multilingual pre-training is essential.** ruBERT (macro F1 = 0.223) fails catastrophically on the code-switching corpus despite the Russian-dominant character of the text. XLM-RoBERTa (F1 = 0.539) and mE5-base (F1 = 0.608) substantially outperform the monolingual baseline, confirming that massively multilingual encoders are a prerequisite for Kazakh-Russian mixed NLP.

2. **LLM few-shot inference outperforms fine-tuned encoders on overall macro F1.** Qwen2.5:7B achieves the highest macro F1 (0.635) and is the only model to detect `STREET_INTERSECTION` entities (F1 = 0.667). However, its throughput (300 samples in 4+ hours) makes it unsuitable for real-time deployment.

3. **Fine-tuned encoders are superior for high-frequency structured entities.** XLM-RoBERTa and mE5-base achieve `ROUTE_NUM` F1 ≈ 0.92, processing thousands of records per minute on GPU. For real-time CTS dispatch triage, XLM-RoBERTa is the recommended production model.

4. `STREET_INTERSECTION` **remains the critical unsolved challenge.** The extreme data scarcity (74 annotated examples) prevents gradient-based models from learning the multi-token conjunctive structure of intersection entities. This is the most important limitation of the current system and the highest-priority target for future data collection.

The contributions of this work are: (1) a custom-annotated bilingual NER corpus for the transport complaint domain, (2) a systematic four-way comparison of NER architectures on Kazakh-Russian code-switching text, (3) entity-level error analysis identifying failure modes specific to this linguistic and domain setting, and (4) practical deployment recommendations for a real-time geotag recognition pipeline.

Future research directions include: targeted data augmentation for rare entities using synthetic data generated from city street graph databases; back-translation between Kazakh and Russian to increase example diversity; hybrid architectures combining a fine-tuned encoder for structured identifiers with an LLM for complex geographic spans; and extension of the pipeline to coordinate resolution through integration with the Astana municipal geocoder API.

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